Lucas Swierad

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Professional Summary

Computer Science and Robotics student at the University of Michigan with hands-on experience in site reliability engineering, full-stack web development, and machine learning. Skilled in cloud infrastructure, CI/CD pipelines, and autonomous systems, with a track record of building and shipping real systems across internships and independent projects.

Education

University of Michigan — Ann Arbor, Michigan
BSE in Computer Science and Robotics.
Relevant Coursework: Data Structures and Algorithms (C++), Intro to Machine Learning (Python), Web Systems, Intro to ROS, Deep Learning for Robotics, Intro to Computer Vision, Kinematics and Dynamics

Experience

Site Reliability Engineering Intern — June 2026 – Present
Barracuda Networks — *Ann Arbor, Michigan*
- Decommissioned 100% of legacy Jenkins infrastructure across Barracuda Networks by migrating 14 active development pipelines to GitHub Actions.
- Secured sensitive pipeline data by integrating managed identities and Azure Key Vault into deployment workflows, automating runtime retrieval of secrets and cryptographic keys to guarantee zero-exposure storage.

Multidisciplinary Design Program Intern — Jan 2026 – Dec 2026
UM-CAEN — *Ann Arbor, Michigan*
- Designing and developing a modular web application to automate administrative workflows for faculty and staff, reducing time spent on calendar management, file organization, and routine notifications.
- Implemented secure backend services using Node.js to support workflow automation, business logic, and API orchestration.
- Integrated Google Workspace APIs (Calendar, Drive, People) and Google OAuth / U-M SSO to enable permission-aware access, calendar automation, and metadata-driven file management.

Teaching Assistant — Jan 2026 – May 2026
UM Robotics — *Ann Arbor, Michigan*
- Prepared instructional materials and led laboratory sections to support student learning across course topics.
- Facilitated lectures and recitation sections, and administered grading and exams to support course assessment.

Projects

AI-Powered Bug Tracker
React.js, FastAPI, MongoDB, Gemini, Python, JavaScript
- Built a responsive frontend using React.js, enabling intuitive data entry, filtering, and visualization of past development experiences.
- Implemented a RESTful backend with FastAPI to manage user data, authentication logic, and API endpoints for structured data retrieval.
- Integrated Google Gemini to provide AI-assisted insights, allowing users to query and summarize historical data to better understand recurring bugs, learning patterns, and technical growth.

Replicating GPT-2
PyTorch, NumPy, Pandas
- Recreated a scaled-down version of OpenAI's GPT-2 architecture from scratch in Python, implementing key transformer components including multi-head attention, positional encoding, and layer normalization.
- Applied linear algebra and deep learning fundamentals to train and fine-tune the model on text datasets, gaining hands-on experience with tokenization, embeddings, and language modeling objectives.

Autonomous Robotic Vehicle
PyTorch, NumPy, OpenCV, ROS
- Developed and evaluated computer vision pipelines for autonomous navigation, including object detection, lane and feature recognition, and environment perception.
- Applied machine learning models to interpret sensor data (camera and IMU), improving robustness of real-time perception under varying environmental conditions.

Skills

Languages: Python, C++, C, JavaScript, SQL

Technologies: Git, GitHub Actions, Azure, Kubernetes, Jenkins, Terraform, AWS, ROS, React.js, Flask, Express.js, PyTorch, Node.js, NumPy, Scikit-Learn, OpenCV, FastAPI, SolidWorks, Onshape